

FARAAZ MOHAMMED

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Full Australian working rights (Temporary Graduate visa, subclass 485) — no sponsorship required • Available immediately

PROFILE

Master of Artificial Intelligence graduate (RMIT, 2026) with a B.Tech in IT and an AI/ML engineering internship shipping production pipelines at national scale. Builds back-end services around models — APIs, request routing, fail-open LLM delegation and evaluation harnesses — and debugs runtime behaviour when systems misbehave. Seeking an AI Software Engineer (Back End) role at Maincode, Melbourne.

TECHNICAL SKILLS

Languages: Python (primary), TypeScript, JavaScript, SQL, Bash, Kotlin

Back end & APIs: Node.js, REST/JSON, MCP servers, Next.js, Prisma/Postgres, service interfaces

AI systems: Ollama, LLM fine-tuning (Unsloth, QLoRA, GGUF), model routing, evaluation design (precision/recall, IoU, sensitivity analysis, ground-truth construction), scikit-learn, XGBoost, OpenCV

Tooling & platforms: Git / GitHub, Docker (basic), Kestra, Linux, Colab / Jupyter, Vercel

AI-assisted engineering: Claude Code (skills, hooks, subagents), Cursor, Anthropic API (prompt caching, routing)

EXPERIENCE

AI/ML Engineering Intern — Prodegee

Mar 2026 – Jun 2026

Remote, Melbourne • RMIT industry capstone, Team DS4 — building-footprint inference for off-grid electrification planning

- Designed and built the school-geolocation algorithm at the core of the team's pipeline (7.2M Overture building polygons, BFS spatial clustering, composite scoring); ran it on 18,121 schools in Côte d'Ivoire and ported it to Burundi and Chad.
- Built an IoU evaluation harness for compound polygons and traced all three generators' failures to a ~35× over-scoped upstream cluster rather than a fitting fault, which redirected the next assignment.
- Built a ±20% sensitivity harness that identified cluster radius as the dominant parameter (±36–44% swing across 191 villages), driving a 75 → 200 m recalibration that fixed a 90%+ household under-count (0.069 → 0.415).
- Led 3 of 10 team assignments and was primary author of 3 of 4 formal deliverables (68 commits, 7 peer reviews); pinned upstream inputs by commit SHA for reproducible re-runs and wrote the semester-2 handoff memo.
- Built a Claude API model router that classifies task complexity with a cheap Haiku call and routes to Haiku / Sonnet / Opus, keeping cost proportional to difficulty.
- Co-built a 40-school satellite-verified ground-truth set and used it to correct a published baseline (6–13% → 37%); published a negative result showing a proximity penalty hurt accuracy at every threshold.
- Replaced a binary distance cutoff with exponential-decay scoring and a weighted confidence score; added per-country sigmoid normalisation so one threshold works in all three countries; authored cross-team handoff contracts.

Tools: Python, Git, Kestra, Docker (basic), scikit-learn, pandas, GeoPandas, Shapely, Claude Code (MCP, hooks)

Dispatch Team Member — Amart Furniture

Feb 2025 – Mar 2026

Melbourne, VIC • part-time, on-site, concurrent with full-time study

- Tracked and reconciled stock in the inventory system, processed customer orders and resolved fulfilment discrepancies across 5 Melbourne stores; trained a new team member.

EDUCATION

Master of Artificial Intelligence — RMIT University

Jul 2024 – Oct 2026

Melbourne, VIC • Coursework: AI Systems Design, Machine Learning, Computer Vision, Intelligent Decision Making (ASP/Clingo)

Bachelor of Technology (Information Technology) — Keshav Memorial Institute of Technology

Dec 2020 – May 2024

Hyderabad, India (affiliated to JNTUH) • Coursework incl. Machine Learning, Neural Networks & Deep Learning, Data Mining

KEY PROJECTS

Gemma Delegation System — local/cloud LLM hybrid • Node.js, MCP SDK, Ollama, Unsloth/QLoRA, Colab

May 2026

- Built an MCP server that lets Claude Code or Cursor delegate lightweight tasks to a locally hosted Gemma model, failing open to the frontier model whenever Ollama is unavailable so delegation never blocks the user.
- Closed the loop: accept/reject feedback updates a capability profile and curates a JSONL dataset; a Colab notebook fine-tunes on confirmed examples (4-bit QLoRA, LoRA r=16) and exports GGUF back into Ollama.

↳ GitHub: github.com/faraaz817/gemma-delegation-system

Bro — interactive AI portfolio chatbot • Next.js, TypeScript, Prisma/Postgres, Anthropic API

2026

- Recruiter-facing site that answers questions from a profile corpus via server-side retrieval with anti-hallucination constraints; Anthropic API, deployed on Vercel.

↳ Live: job-go-mauve.vercel.app/portfolio

Fire Detection & Alerting System — B.Tech capstone, team of 4 · Python, OpenCV

2024

- Trained a custom OpenCV Haar cascade for real-time fire detection, chosen over heavier detectors for latency; a threaded pipeline raises an alarm and sends WhatsApp alerts to the nearest fire station.

Time Flow — ADHD-focused planner (PWA) · JavaScript, Service Worker

Jul – Sep 2026

- Built and deployed an installable, offline-first PWA (4,600 lines of vanilla JS); fixed input-focus loss under 1-second live re-renders with a targeted DOM-patch path instead of a full re-render.
- Implemented dual-timer focus sessions, weekly/monthly reminder scheduling and a cache-first service worker for system notifications; built a JS ↔ Kotlin bridge so the same code runs in a native Android wrapper with exact alarms.

↳ Live: faraaz817.github.io/time-flow-web · GitHub: github.com/faraaz817/time-flow-web

MGM Prayer Alerts — Android app for Melbourne Grand Mosque · Kotlin, AlarmManager, JUnit

Aug 2026

- Wrote functional requirements before coding, then fixed a critical AM/PM storage bug and a refresh alarm that stopped re-arming; unit-tested time parsing. In use by community members.

↳ GitHub: github.com/faraaz817/MgmPrayerAlerts

PUBLICATION

K. M. Asudaria, **M. F. Abdul Khadeer**, S. S. Rafai, K. S. Reddy. “A Comparative Study of Edge Detection Techniques to Identify Maize Leaf Diseases using Machine Learning.” *Quest Journal of Software Engineering and Simulation*, Vol. 9(12), pp. 43–51, Dec 2023 (ISSN 2321-3795, peer-reviewed). Benchmarked 5 edge-detection operators × 8 classifiers; best result Sobel + MLP at 94.75%. Implemented the Canny arm (OpenCV + scikit-learn). ↳ questjournals.org/jses · Code: github.com/faraaz817/Machine_Learning

CERTIFICATIONS, AWARDS & LEADERSHIP

- IBM Z Xplore – Concepts and IBM Z & LinuxONE Community Contributor – Level 1 (IBM digital badges via Credly, 2026).
- Smart India Hackathon delegate, Government of India (2022) · Certificate of Publication, Quest Journals (2023).
- Head, Vachan Speakers Club, KMIT (2022–23) — led public-speaking workshops, debates and communication seminars.
- Volunteer fundraiser, Taher Foundation (2023).

LANGUAGES

English (native / bilingual) • Urdu (native / bilingual) • Hindi (full professional) • Telugu (elementary)

REFEREES

Available on request.